



PUBLIC UTILITIES REGULATORY COMMISSION

2023 First Quarter Natural Gas, Electricity & Water Tariff Decision

An In-depth Look at the Rationale
Underpinning the 2023 First Quarter
Natural Gas, Electricity and Water Tariffs

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1.0 INTRODUCTION

As part of the Public Utilities Regulatory Commission's strategy to ensuring the real value of natural gas, electricity and water tariffs as approved by the Commission and payable by consumers at any particular point in time, the Commission has put in place Guidelines as a governance framework for Quarterly Review of foregoing tariffs. The legal foundation and philosophy of the Guidelines are rooted in a number of statutory provisions as captured in the Public Utilities Regulatory Commission Act 1997, (Act 538). These statutory provisions as captured in the PURC Act are summarized as follows.

1. Protection of Consumer interest - Section 16 (3)(a)
2. Assurance of Investor / Utility interest- Section 16(3)(b); 3(c)
3. Assuring reasonable cost of production of the service - Section 16(3)(c)
4. Assurance of the financial viability of the Public Utility - Section 16(3)(d)
5. Uniformity of prices throughout the country - Section 20(1)
6. Best use of natural resources - Section 20(1)(b)
7. Economic development of the country

- Section 20(1)(c)

8. Different rates for different consumer classes - Section 20(2)

Within above legal mandates, the main objective of the Commission's Rate Setting Guidelines for Quarterly Review of Natural Gas, Electricity and Water Tariffs is to reflect the effect of changes in macroeconomic and market-driven factors in the cost of operations of the Natural Gas, Electricity and Water Utility Service Providers (USPs) as well as Consumers, since the effects of these factors are beyond the control of the USPs.

Thus, pursuant to section 2.4 of PURC's Rate Setting Guidelines for Electricity Distribution and Supply, the Commission carried out a review of Natural Gas, Electricity and Water Tariffs for the First Quarter of 2023.

This decision paper highlights the results/ findings in terms of the impact of changes in actual and projected Hydro-Thermal Generation Mix, Ghana Cedi-US Dollar Exchange Rate, Inflation and Fuel Prices, as well as the Commission's decisions thereof for the First Quarter of 2023 effective February 01, 2023.

2.0 METHODOLOGY, ANALYSIS AND FINDINGS

2.1 Methodology

The methodology employed in determining the effect of the various variables involves a comparative analysis of actual and projected Hydro-Thermal Generation Mix, Ghana Cedi-US Dollar Exchange Rate, Inflation Rate and Fuel Prices, which data were used in determination of tariffs for the 2022-2025 Multi-Year Major Tariff Review. Under this framework, projections were made for Exchange Rate based on Actual Inter-Bank Ghana Cedi-US Dollar Selling Exchange Rate

from data from Bank of Ghana, Inflation Rate for 2022 from Ghana Statistical Service and Natural Gas Prices from various Gas suppliers as well as Projected Hydro Electrical Energy Generation from both Akosombo and Kpong generating stations data submitted by the Volta River Authority to the Electricity Market Oversight Panel (EMOP) and PURC, whilst accounting for actual data in respect of above variables so as to address under and over recovery of the tariff which may have occurred over previous quarters as approved by the Board.

2.2 Analysis

To determine the direction of the Commission's decision with respect to 2023 First Quarter tariff results, underpinning data with respect to Generation mix (Hydro-Thermal Mix), Ghana Cedi-US Dollar Exchange Rate, Inflation Rate and Natural Gas Price were analysed using

the methodological framework noted in the previous section. The selected data which were analysed are presented in Table-1.

2.2.1 Underpinning Data

The underpinning data of the 2023 First Quarter Tariff analysis are summarised and presented in Table-1.

Table 1: Summary of Data Used in Analysis of 2023 First Quarter Electricity Tariffs

Item No.	Item Description	Unit	2022-2025 Parameters/Assumptions Effective September 01 2022	Q1 2023 Parameters/Assumptions Effective February 01, 2023
A.	Generation Mix:			
A1	Hydro	%	20.20	26.11
A2	Thermal	%	79.80	73.89
B.	Exchange Rate:			
	PURC Projected	GHS/USD ExRate	7.5165	10.5421
C.	Inflation:			
	Projected Annual Average	%	10.06	42.63
D.	Fuel Price :			
	Natural Gas	US\$/MMBtu	5.9060	6.0952

Source: PURC Data & Tariff Analysis, 2023

2.3 Findings/Results from Analysis of Data

The findings/results from analysis of Data Captured in Table-1 are discussed as follows.

2.3.1 Hydro -Thermal Electrical Energy Generation Mix

The data presented in Table 1 with respect to Hydro-Thermal generation indicates the following:

- The hydro generation mix of 26.11% represents hydro generation as a percentage of total electrical energy generated from both hydro and thermal sources, indicating a 5.91% variance on 2022-2025 projected (existing). In a nutshell, this represents 29.3% increase over the hydro generation mix used in 2022-2025 Multi-Year Major Tariff Review.

- The projected thermal generation mix of 73.89% represents a 7.40% reduction over 2022-2025 Multi-Year Major Tariff Review projected thermal generation of 79.80%. This was as a result of increase in 2023 projected hydro electrical energy generation from Akosombo and Kpong by VRA to the regulated electricity market.

2.3.2 Ghana Cedi-US Dollar Exchange Rate

With respect to Ghana Cedi-US Dollar Exchange Rate the data shown in Table-1 indicates the following:

- A projected Weighted Average Ghana Cedi-US Dollar Exchange Rate of GHS10.5421/US\$1.0000. This projection is based on Actual Inter-Bank Average Ghana Cedi-US Dollar Selling Exchange Rate for the period September-December, 2022.

- b) This Rate represents a projected 40.25% depreciation of the Ghana Cedi against the US Dollar over 2022-2025 Multi-Year Major Tariff Review Exchange Rate of GHS7.5165/US\$1.0000.

2.3.3 Inflation Rate

Similar to data and approach in respect of Ghana Cedi-US Dollar Exchange Rate noted in the previous section, a projected Ghana Average Inflation Rate of 42.63% was used for the First Quarter Of 2023. This represents Ghana's average projected inflation rate variance of 32.57% between 2022-2025 Multi-Year Major Tariff Review Projected Average Inflation Rate of 10.06% and the Projected Average Inflation Rate noted above.

2.3.4 Price of Fuel - Natural Gas

The two primary fuels used in determination of Fuel Recovery Charge as a component of the individual power plant tariffs hence Composite Bulk Generation Tariff for the 2022-2025 Multi-Year Major Tariff Review

were Natural Gas and Heavy Fuel Oil (HFO). However, for First Quarter 2023, no power plant is expected to use HFO due to AKSA's conversion to gas usage. With regards to Natural Gas, the Commission approved a Weighted Average Cost of Gas (WACOG) of US\$6.0972/MMBtu which is an increase of 3.20% over the 2022-2025 Multi-Year Major Tariff Review Existing WACOG of US\$5.9060/MMBtu. The increase in WACOG is mainly as a result of change in Gas Commodity Prices due to individual escalation clauses in contracts in addition to under and over recovery of revenue due to actual gas supplied volume varying from projected gas volumes used in determination of 2022-2025 Multi-Year Major Tariff Review WACOG

It must be noted that though Ghana's dominant fuel is now natural gas, there is not much benefit in terms of reduction in tariffs for the quarter based on Natural Gas Fuel Recovery Charge due to high priced long term domestic Natural Gas contracts.

3.0 SUMMARY OF ELECTRICITY TARIFF RESULTS

Taking into consideration the various assumptions and projections noted in Section 2, tariff results from analyses of data discussed above are summarised and presented in Table-2.

Table 2: Summary of Tariff Results for 2022 Fourth Quarter

Item No.	Item Description	Measure	2022-2025 Tariff Results Effective September 01 2022	Q1 2023 Tariff Results Effective February 01, 2023
A.	Generation Tariffs:			
A1	VRA Bulk Generation Charge (BGC)	GHp/kWh	37.9560	44.4570
A2	Composite Bulk Generation Charge (VRA&IPP)	GHp/kWh	63.1997	85.8447
B.	Transmission Tariffs:			
B1	Transmission Service Charge Attributable to Network Business (TSC-1)	GHp/kWh	7.9090	8.4442
B2	Transmission Service Charge Attributable to Losses (TSC-2)	GHp/kWh	2.7020	3.6701
C.	Distribution Tariffs:			
C1	Distribution Service Charge Attributable to Network Business (DSC-1)	GHp/kWh	17.0429	18.3337
C2	Distribution Service Charge Attributable to Losses (DSC-2)	GHp/kWh	18.2277	24.9450

3.1 Bulk Generation Charge

Projecting for First Quarter 2023, VRA Bulk Generation Tariff increased from GHp37.9560/kWh to GHp44.4570/kWh while the Composite Bulk Generation Tariff (VRA&IPP) increased from GHp63.1997/kWh to GHp85.8447/kWh representing a 17.13% and 35.83% increase in VRA Bulk Generation Tariff and Composite Generation Tariff respectively. The change is as a result upward movement in Ghana Cedi-US Dollar Exchange Rate and Hydro-Thermal Generation Mix.

3.2 Transmission Service Charge

The Transmission Tariff attributable to the network business (TSC-1) increased from GHp7.9090/kWh to GHp8.4442/kWh in First Quarter, 2023 while that of losses (TSC-2) increased from GHp2.7020/kWh to GHp3.6701/kWh representing 6.77% and 35.83% increase respectively. The change in TSC-1 is as a result upward movement in Ghana Cedi-US Dollar Exchange Rate and Inflation Rate whilst the change in TSC-2 is as a result of result rise in Ghana Cedi-US Dollar Exchange Rate and variation in Hydro-Thermal Generation Mix.

3.3 Distribution Service Charge

With respect to Distribution Service Charge (DSC) for the First Quarter of 2023, the tariff attributable to the network business (DSC-1) increased from GHp17.0429/kWh to GHp18.3337/kWh while losses (DSC-2) increased from GHp18.2277/kWh to GHp24.9450kWh. The upward adjustment in DSC-1 is as a result rise in Ghana Cedi-US Dollar Exchange Rate and Inflation Rate. The DSC-2 on the other hand was affected by the movement in Ghana Cedi-US Dollar Exchange Rate and variation in Hydro-Thermal Generation Mix.

3.4 Impact of Changes in Composite Bulk Generation Charge, Transmission Service Charge and Distribution Service Charge on End User Tariffs

The impact of changes in the Composite Bulk Generation Tariff, Transmission and Distribution Value Added on End User Tariffs as indicated in Table-2 and related sections are presented in this section based on the effect of each variable on rates payable by consumers.

3.4.1 Generation Mix Effect

With regards to Generation Mix effect, holding Ghana Cedi-Us Dollar Exchange, Inflation Rate and Natural Gas Price constant, Revenue Gap Analysis indicates GHS129.35 million revenue over-recovery resulting from increase in projected hydro generation for 2023 First Quarter beyond the base projected for 2022-2025 Multi-Year Major Tariff Review.

3.4.2 Exchange Rate Effect

With respect to Ghana Cedi-US Dollar Exchange effect, holding Generation Mix, Inflation Rate and Natural Gas Price constant at 2022-2025 Multi-Year Major Tariff Review values, Revenue Gap Analysis indicates a GHS1.12 billion revenue under-recovery by Utility Service Providers resulting from a 40.25% loss in value of the Ghana Cedi against the US Dollar.

3.4.3 Inflation Rate Effect

In terms of Inflationary effect, holding Ghana Cedi-Us Dollar Exchange, Generation Mix and HFO Price constant, Revenue Gap Analysis shows a GHS35.65 million revenue under-recovery resulting from a jump in the Inflation Rate from 10.06% to an average Inflation Rate of 42.63%.

3.4.3 Natural Gas Price Effect

Natural Gas Price effect was analysed from the perspective of upward movement in the projected WACOG from USD 5.9060/MMBtu to US\$6.0972/MMBtu and conversion of AKSA's conversion to gas usage holding Ghana Cedi-US Dollar Exchange, Inflation Rate and Generation Mix effect constant. Revenue Gap Analysis indicates a GHS39.08 million revenue under-recovery.

3.5 Combined Effect of Generation Mix, Ghana Cedi-US Dollar Exchange, Inflation Rate and Natural Gas Price on Electricity Tariffs Payable by Consumers

The Hydro-Thermal Generation Mix, Ghana Cedi-US Dollar Exchange, Inflation Rate and Natural Gas Price were identified from 2023 First Quarter Tariff analysis as having effects on tariffs payable by consumers. The combined impact of these four effects indicates a total amount of GHS1.10 billion recoverable from consumers over 2023 First Quarter, representing an increase of 29.96% across board in electricity tariffs payable by all categories of consumers.

4.0 SUMMARY OF WATER TARIFF RESULTS

With regards to quarterly review of water tariffs, the Commission took into consideration the Ghana Cedi-US Dollar Exchange, Inflation Rate and Cost of Electricity as key variables impacting on water tariffs payable by consumers for First Quarter of 2023. The effect of change in each of the aforementioned variables on Water Tariffs payable by consumers are presented in this section.

4.1 Electricity Price Effect

The cost of electricity, it must be stated, constitutes a significant portion of Ghana Water Company Limited (GWCL) total cost of water production, transmission and distribution. With respect to Electricity Price effect, holding Ghana Cedi-US Dollar Exchange and Inflation Rate constant, Revenue Gap Analysis shows GHS 27.60 million revenue under-recovery resulting from increase in projected electricity cost for 2023 First Quarter beyond the base projected for 2022-2025 Multi-Year Major Tariff Review.

4.2 Exchange Rate Effect

With regards to Ghana Cedi-US Dollar Exchange effect, holding Electricity Price Effect and Inflation Rate constant at 2022-

2025 Multi-Year Major Tariff Review values, Revenue Gap Analysis indicates a GHS39.77 million revenue under-recovery by GWCL resulting from a 40.25% loss in value of the Ghana Cedi against the US Dollar.

4.3 Inflation Rate Effect

Similar to above, holding Electricity Price Effect and Ghana Cedi-US Dollar Exchange constant, Revenue Gap Analysis shows a GHS13.09 million revenue under-recovery resulting from a rise in the 2022-2025 Multi-Year Major Tariff Review Average Inflation Rate from 10.06% to an average Inflation Rate of 42.63%.

4.4 Combined Effect of Ghana Cedi-US Dollar Exchange, Inflation Rate and Electricity Tariffs on Water Tariffs Payable by Consumers

The combined impact of these three effects indicates a total amount of GHS80.46 million recoverable from consumers over 2023 First Quarter. The commission approved varying increases with majority of consumers experiencing an 8.30% increase in tariffs over existing water tariffs.

5.0 COMMISSION'S DECISION ON 2023 FIRST QUARTER TARIFFS

In light of results from analysis of data discussed in previous sections, the Commission has taken the following decisions.

1. The Commission approved a 29.96% across board increase in electricity tariffs payable by consumers.
2. The Commission undertook a restructuring of GWCL's consumer categories to enable the company meet consumer demand in terms of provision of water services over the First Quarter of 2023. In that regard, water tariffs were increased by 8.30% with other consumers experiencing varying percentage adjustments.
3. To address the financial challenges facing Natural Gas, Electricity and Water Utility Service Providers, the Commission encourages doubling of efforts in revenue collection by Utility Service Providers.
4. The Commission will like to assure all Stakeholders that it will continue to strictly implement its Quarterly Tariff Review per its Rate Setting Guidelines for Quarterly Review of Natural Gas, Electricity and Water Tariffs in order to address changes in operational conditions whilst holding the Utility Service Providers to strict adherence to its regulatory standards and benchmarks by continually monitoring operations of the Utility Service Providers to ensure value for money and quality of service delivery.

APPENDIX 1: APPROVED ELECTRICITY TARIFFS EFFECTIVE FEBRUARY 01, 2023

CUSTOMER CLASS	APPROVED TARIFF (GHP/KWH)
Residential	
0-30	54.4627
51-300	115.7212
301-600	150.1837
600+	166.8708
Service Charge for Lifeliners	213.0000
Service Charge for Other Residential Consumers	1073.0886
Non-Residential	
0-300	108.8876
301-600	115.8681
600+	172.9692
Service Charge	1242.8245
SLT-LV	
Energy Charge (GHP/kWh)	172.3461
Service Charge (GHP/month)	50000.00
SLT-MV	
Energy Charge (GHP/kWh)	130.8541
Service Charge (GHP/month)	50000.00
SLT-HV	
Energy Charge (GHP/kWh)	137.3370
Service Charge (GHP/month)	50000.00
SLT-HV STEEL COMPANIES	
Energy Charge (GHP/kWh)	96.8627
Service Charge (GHP/month)	50000.00
SLT-HV Mines	
Energy Charge (GHP/kWh)	343.0618
Service Charge (GHP/month)	50000.00

APPENDIX 2: APPROVED WATER TARIFFS EFFECTIVE FEBRUARY 01, 2023

TARIFF CATEGORY		APPROVED TARIFF
Residential		
0-5m ³ (Exclusive)	- GHp/1000 Litres	433.3840
Above 5 m ³	- GHp/1000 Litres	737.4737
Service Charge	- GHp/Month	1000.00
Non-Residential		
	- GHp/1000 Litres	1215.1142
Service Charge	- GHp/Month	2000.00
Sachet Water Producers		
	- GHp/1000 Litres	1800.00
Service Charge	- GHp/Month	10000.00
Commercial Bottled Water and Drinks		
	- GHp/1000 Litres	3000.00
Service Charge	- GHp/Month	25000.00
Other Commercial		
	- GHp/1000 Litres	3000.00
Service Charge	- GHp/Month	10000.00
Industrial		
	- GHp/1000 Litres	2000.00
Service Charge	- GHp/Month	50000.00
Public Institutions/Government Departments		
	- GHp/1000 Litres	946.0676
Service Charge	- GHp/Month	2000.00
Public Stand Pipes		
	- GHp/1000 Litres	486.3528
Service Charge	- GHp/Month	1000.00
Ports and Harbours		
	- GHp/1000 Litres	3000.00
Service Charge	- GHp/Month	100000.00
Bulk Supply		
	- GHp/1000 Litres	433.3840
Service Charge	- GHp/Month	10000.00

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